Chapter 11: Permutations, Combinations, and the Binomial Theorem

MULTIPLE CHOICE

1. What is the value of 6!?

A. 46 656
B. 720
C. 21
D. 36

ANS: B DIF: Easy OBJ: Section 11.1 NAT: PC2
TOP: Permutations KEY: factorial

2. The number of different ways that 9 bikes can be locked in a bike rack is

A. 3 628 800
B. 20 160
C. 40 320
D. 362 880

ANS: D DIF: Easy OBJ: Section 11.1 NAT: PC1
TOP: Permutations KEY: fundamental counting principle

3. Evaluate ${}_{15}P_8$.

A. 32 432 400
B. 163 459 296 000
C. 6435
D. 259 459 200

ANS: D DIF: Easy OBJ: Section 11.1 NAT: PC2
TOP: Permutations KEY: permutations

4. Solve for the variable:
${}_5P_r = 20$

A. 5
B. 2
C. 60
D. 7

ANS: B DIF: Easy OBJ: Section 11.1 NAT: PC2
TOP: Permutations KEY: permutations

5. Solve for the variable:
${}_nC_5 = 252$

A. 15
B. 5
C. 6
D. 10

ANS: D DIF: Difficult OBJ: Section 11.2 NAT: PC3
TOP: Combinations KEY: combinations

6. An orchestra has 2 violinists, 3 cellists, and 4 harpists. Assume that the players of each instrument have to sit together, but they can sit in any position in their own group. In how many ways can the conductor seat the members of the orchestra in a line?

A. 144
B. 72
C. 24
D. 1728

ANS: D DIF: Average OBJ: Section 11.1 NAT: PC2
TOP: Permutations KEY: fundamental counting principle

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7. Evaluate the expression ${}_{14}C_9$.

A. 726 485 760
C. 2002
B. 240 240
D. 126

ANS: C DIF: Easy OBJ: Section 11.2 NAT: PC3
TOP: Combinations KEY: combinations

8. Evaluate the expression ${}_{14}C_{11} \times {}_{14}P_3$.

A. 2184
C. 794 976
B. 87 178 291 200
D. 239 500 800

ANS: C DIF: Average OBJ: Section 11.1 | Section 11.2
NAT: PC2 | PC3 TOP: Permutations | Combinations KEY: permutations | combinations

9. Solve for *n* in the expression ${}_nC_4 \times {}_nP_4 = 117\,600$.

A. 8
C. 9
B. 16
D. 8

ANS: D DIF: Average OBJ: Section 11.1 | Section 11.2
NAT: PC2 | PC3 TOP: Permutations | Combinations KEY: permutations | combinations

10. After the tryouts for the volleyball team, the coach selects 14 people to join the team. Due to a problem with transportation, only 9 people can travel. In how many ways can the coach pick the people to go?

A. 726 485 760
C. 630
B. 2002
D. 126

ANS: B DIF: Average OBJ: Section 11.2 NAT: PC3
TOP: Combinations KEY: combinations

11. Rachael has a digital music player that holds 800 songs. She has 1500 songs in her music library. She decides that her 50 favourite songs must be on the player. Which expression can be used to calculate the number of ways can she load songs on to the MP3 player so that it is full?

A. ${}_{800}C_{50}$
C. ${}_{1500}C_{50}$
B. ${}_{1450}C_{750}$
D. ${}_{1500}C_{800}$

ANS: B DIF: Average OBJ: Section 11.2 NAT: PC3
TOP: Combinations KEY: combinations

12. For a mock United Nations, 6 boys and 7 girls are to be chosen. If there are 12 boys and 9 girls to choose from, how many groups are possible?

A. 846 720
C. 960
B. 33 264
D. 120 708 403 200

ANS: B DIF: Difficult OBJ: Section 11.2 NAT: PC3
TOP: Combinations KEY: combinations

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13. Jenni and Hari go to a local Chinese restaurant for dim sum. If there are 20 items on the menu, and Jenni orders 7 items and Hari orders 11 items, which expression represents the total number of choices between them?

A. $(11!)(7!)$

B. ${}_{20}C_7 \times {}_{20}C_{11}$

C. ${}_{20}P_7 \times {}_{20}P_{11}$

D. $\frac{20!}{9!13!}$

ANS: B DIF: Difficult OBJ: Section 11.2 NAT: PC3
TOP: Combinations KEY: combinations

14. Determine the coefficient, a, for the term ax^5y^7 of the binomial expansion of $(x+y)^{12}$.

A. 35

B. 60

C. 792

D. 3 991 680

ANS: C DIF: Average OBJ: Section 11.3 NAT: PC4
TOP: The Binomial Theorem KEY: binomial expansion | binomial theorem

15. For which of the following terms is $a = 55$ in the expansion of $(x + y)^{11}$?

A. ax^3y^8

B. ax^8y^3

C. ax^2y^9

D. ax^{11}

ANS: C DIF: Average OBJ: Section 11.3 NAT: PC4
TOP: The Binomial Theorem KEY: binomial expansion | binomial theorem

16. Determine the 5th term in the expansion of $(x-2)^{12}$.

A. $-7920x^8$

B. $-7920x^5$

C. $202\,752x^{12}$

D. $7920x^8$

ANS: D DIF: Difficult OBJ: Section 11.3 NAT: PC4
TOP: The Binomial Theorem KEY: binomial expansion | binomial theorem

17. Prom's friend gives him a row of Pascal's triangle and asks which row it comes from. Prom adds the numbers and obtains a sum of 65 536. Which row do the numbers come from?

A. 34

B. 18

C. 17

D. 16

ANS: C DIF: Difficult OBJ: Section 11.3 NAT: PC4
TOP: The Binomial Theorem KEY: Pascal's triangle

18. How many terms are there in the expansion of $(2x-4y)^{12}$?

A. 6

B. 13

C. 12

D. 11

ANS: B DIF: Average OBJ: Section 11.3 NAT: PC4
TOP: The Binomial Theorem
KEY: Pascal's triangle | binomial expansion | binomial theorem

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19. Which of the following has 30 terms in its binomial expansion?

A. $(-2y)^{29}$ **C.** $(-4x-2y)^{30}$

B. $(-4x-2y)^{29}$ **D.** $(-4x-2y)^{31}$

ANS: B DIF: Average OBJ: Section 11.3 NAT: PC4
TOP: The Binomial Theorem
KEY: Pascal's triangle | binomial expansion | binomial theorem

20. A scout troop is arranged in a circle for an opening ceremony. If there are 9 scouts, in how many unique ways can they stand around the circle?

A. 5040 **C.** 81

B. 40 320 **D.** 362 880

ANS: B DIF: Difficult OBJ: Section 11.1 NAT: PC1
TOP: Permutations KEY: factorial | fundamental counting principle

21. A scout troop is arranged in a circle for an opening ceremony with its leaders. Assuming there are 7 troop leaders who stand together for the ceremony and 39 scouts in total, including the troop leaders, in how many unique ways can they be arranged around the circle?

A. $(32!)(7!)$ **C.** $(39!)(32!)$

B. ${}_{39}C_{32}$ **D.** ${}_{39}C_{7}$

ANS: A DIF: Difficult + OBJ: Section 11.1 NAT: PC1
TOP: Permutations KEY: factorial | fundamental counting principle

22. Suppose that 10 non-collinear points are plotted on a plane. Which expression represents the number of triangles that can be formed using these points?

A. ${}_{10}P_{7}$ **C.** ${}_{10}C_{3}$

B. $\frac{10!}{3!}$ **D.** $\frac{10!}{(10-3)!}$

ANS: C DIF: Average OBJ: Section 11.2 NAT: PC3
TOP: Combinations KEY: combinations

23. The leadership committee at a high school has 4 grade 10 students, 2 grade 11 students, and 6 grade 12 students. This year, 12 grade 10, 8 grade 11, and 10 grade 12 students applied for the committee. How many ways are there to select the committee?

A. 2 910 600 **C.** 733

B. 100 590 336 000 **D.** 163 136

ANS: A DIF: Difficult OBJ: Section 11.2 NAT: PC3
TOP: Combinations KEY: combinations | fundamental counting principle

24. After a sports, tournament every player shakes hands with every other player once. If there are 36 handshakes in total, how many players are at the tournament?

A. 18 **C.** 8

B. 10 **D.** 9

ANS: D DIF: Difficult + OBJ: Section 11.2 NAT: PC3
TOP: Combinations KEY: combinations

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25. While at the dollar store, Peter finds 19 items at $1 each that he wants, but he only has $3. The number of ways he could select which items to buy is

A. ${}_{19}C_0 + {}_{19}C_1 + {}_{19}C_2 + {}_{19}C_3$

B. $({}_{19}C_0)({}_{19}C_1)({}_{19}C_2)({}_{19}C_3)$

C. $({}_{19}P_0)({}_{19}P_1)({}_{19}P_2)({}_{19}P_3)$

D. ${}_{19}P_0 + {}_{19}P_1 + {}_{19}P_2 + {}_{19}P_3$

ANS: A DIF: Difficult OBJ: Section 11.2 NAT: PC3
TOP: Combinations KEY: combinations

COMPLETION

1. The selection of objects in which order is not important is called a(n) _____.

ANS: combination

DIF: Easy OBJ: Section 11.1 | Section 11.2 | Section 11.3
NAT: PC1 | PC2 | PC3 | PC4
TOP: Permutations | Combinations | The Binomial Theorem
KEY: combinations | factorial | permutations | Pascal's triangle

2. ${}_{14}P_9$ can be written in factorial notation as _____.

ANS: $\frac{14!}{(14-9)!}$

DIF: Easy OBJ: Section 11.1 NAT: PC2 TOP: Permutations
KEY: permutations | factorial

3. ${}_{20}C_{14}$ can be written in factorial notation as _____.

ANS: $\frac{20!}{(20-14)!14!}$

DIF: Easy OBJ: Section 11.2 NAT: PC3 TOP: Combinations
KEY: combinations | factorial

4. The expansion of the binomial $(x+y)^{10}$ includes the coefficients in row _____ of Pascal's triangle.

ANS: 11

DIF: Easy OBJ: Section 11.3 NAT: PC4 TOP: The Binomial Theorem
KEY: binomial expansion | binomial theorem | coefficient

5. The sum of the numbers in the 13th row in Pascal's triangle is _____.

ANS: 4096

DIF: Average OBJ: Section 11.3 NAT: PC4 TOP: The Binomial Theorem
KEY: Pascal's triangle

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MATCHING

Match each value with the correct expression.

A. ${}_7C_4$

B. ${}_{11}C_7$

C. ${}_7P_4$

D. ${}_{11}P_7$

E. $\frac{7!}{4!}$

F. $\frac{11!}{7!}$

1. 330
2. 840
3. 1 663 200
4. 210
5. 35

1. ANS: B DIF: Average OBJ: Section 11.1 | Section 11.2 | Section 11.3
NAT: PC1 | PC2 | PC3 TOP: Permutations | Combinations
KEY: permutations | combinations | factorial

2. ANS: C DIF: Average OBJ: Section 11.1 | Section 11.2 | Section 11.3
NAT: PC1 | PC2 | PC3 TOP: Permutations | Combinations
KEY: permutations | combinations | factorial

3. ANS: D DIF: Average OBJ: Section 11.1 | Section 11.2 | Section 11.3
NAT: PC1 | PC2 | PC3 TOP: Permutations | Combinations
KEY: permutations | combinations | factorial

4. ANS: E DIF: Average OBJ: Section 11.1 | Section 11.2 | Section 11.3
NAT: PC1 | PC2 | PC3 TOP: Permutations | Combinations
KEY: permutations | combinations | factorial

5. ANS: A DIF: Average OBJ: Section 11.1 | Section 11.2 | Section 11.3
NAT: PC1 | PC2 | PC3 TOP: Permutations | Combinations
KEY: permutations | combinations | factorial

SHORT ANSWER

1. Josie goes to the local submarine sandwich shop. She can choose from 21 types of sandwiches. There are 3 types of cheese, 8 types of vegetables, and 7 types of sauce. Assuming that Josie only chooses one type of cheese, one vegetable, and one sauce for her sandwich, how many choices does she have?

ANS:
Let s represent the number of sandwich choices, c the number of cheese choices, v the number of vegetable choices, and a the number of sauce choices. Let T represent the total number of choices.

$T = (s)(c)(v)(a)$

$= (21)(3)(8)(7)$

$= 3528$

There are 3528 different choices.

DIF: Average OBJ: Section 11.1 NAT: PC1 TOP: Permutations
KEY: fundamental counting principle

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2. Sun Eui and Pattie are arguing about the number of ways the letters in PERMUTATION and COMBINATION can be rearranged. Sun Eui believes that there are more ways to rearrange PERMUTATION, while Pattie thinks that COMBINATION has more unique arrangements. Which person is correct?

ANS:
Let P represent the number of ways the letters in PERMUTATION can be arranged, and C the number of ways the letters in COMBINATION can be arranged.

$$P = \frac{11!}{2!}$$

$$= 19\,958\,400$$

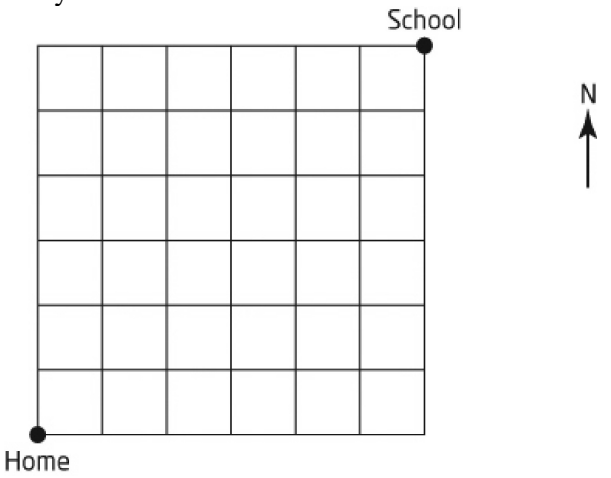
$$C = \frac{11!}{2!2!2!}$$

$$= 4\,989\,600$$

There are more arrangements for PERMUTATION, so Sun Eui is correct.

DIF: Average OBJ: Section 11.1 NAT: PC1 TOP: Permutations
KEY: fundamental counting principle

3. Joe wants to travel from his home to school. The school is 6 blocks east and 6 blocks north. How many routes can Joe take from his house to school if he only moves east and north.



ANS:
The number of different routes can be calculated using the following permutation:

$$\frac{12!}{6!6!} = 924$$

Joe can take one of 924 different routes to travel from home to school.

DIF: Average OBJ: Section 11.1 NAT: PC1 TOP: Permutations
KEY: permutations

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4. Show that ${}_{12}C_5 = {}_{12}C_7$.

ANS:

$$\textbf{L.S.} = \frac{12!}{(12-5)!5!} \qquad \textbf{R.S.} = \frac{12!}{(12-7)!7!}$$

$$= \frac{12!}{7!5!} \qquad = \frac{12!}{5!7!}$$

$$= \frac{12!}{7!5!}$$

L.S. = R.S.

DIF: Average OBJ: Section 11.2 NAT: PC2 TOP: Combinations
KEY: combinations

5. Use the binomial theorem to expand $\left(\frac{a}{2} - \frac{b}{3}\right)^4$.

ANS:

$$\left(\frac{a}{2} - \frac{b}{3}\right)^4 = {}_4C_0\left(\frac{a}{2}\right)^4\left(-\frac{b}{3}\right)^0 + {}_4C_1\left(\frac{a}{2}\right)^3\left(-\frac{b}{3}\right)^1 + {}_4C_2\left(\frac{a}{2}\right)^2\left(-\frac{b}{3}\right)^2 + {}_4C_3\left(\frac{a}{2}\right)^1\left(-\frac{b}{3}\right)^3 + {}_4C_4\left(\frac{a}{2}\right)^0\left(-\frac{b}{3}\right)^4$$

$$= 1\left(\frac{a}{2}\right)^4\left(-\frac{b}{3}\right)^0 + 4\left(\frac{a}{2}\right)^3\left(-\frac{b}{3}\right)^1 + 6\left(\frac{a}{2}\right)^2\left(-\frac{b}{3}\right)^2 + 4\left(\frac{a}{2}\right)^1\left(-\frac{b}{3}\right)^3 + 1\left(\frac{a}{2}\right)^0\left(-\frac{b}{3}\right)^4$$

$$= \frac{a^4}{16} - \frac{a^3b}{6} + \frac{a^2b^2}{6} - \frac{2ab^3}{27} + \frac{b^4}{81}$$

DIF: Average OBJ: Section 11.3 NAT: PC4 TOP: The Binomial Theorem
KEY: binomial expansion | binomial theorem

6. There are 5 men and 8 women available for a 5-person committee. If the committee must have at least one man and one woman, how many possible committees are there?

ANS:
The total number of possible committees is ${}_{13}C_5 = 1287$.
The total number of committees with no men is ${}_8C_5 = 56$. The total number of committees with no women is ${}_5C_5 = 1$.
The total number of committees with at least one man and one woman is 1287 – 56 – 1 or 1230.

DIF: Average OBJ: Section 11.2 NAT: PC3 TOP: Combinations
KEY: combinations

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7. Simplify the expression $\frac{(2n+2)!}{(2n-2)!0!}$.

ANS:

$$\frac{(2n+2)!}{(2n-1)!0!} = \frac{1 \cdot 2 \cdot 3 \cdot \ldots \cdot (2n-2)(2n-1)(2n)(2n+1)(2n+2)}{1 \cdot 2 \cdot 3 \cdot \ldots \cdot (2n-2)(2n-1) \cdot 1}$$

$$= (2n)(2n+1)(2n+2)$$

$$= 2n(4n^2 + 6n + 2)$$

$$= 8n^3 + 12n^2 + 4n$$

DIF: Average OBJ: Section 11.1 NAT: PC1 TOP: Permutations
KEY: factorial

8. A neon sign with the words "Espresso Coffee" on it has 5 letters burnt out. In how many ways can you select 3 good letters and 2 burnt-out letters?

ANS:
There are 14 letters in total. 5 of these are burnt out.

$$({}_9C_3)({}_5C_2) = 84 \times 10$$

$$= 840$$

There are 840 ways to select 3 good letters and 2 burnt-out letters.

DIF: Average OBJ: Section 11.1 NAT: PC3 TOP: Combinations
KEY: combinations

9. A math teacher is preparing a quiz for all of the students in grade 12. She wants to give each student the same questions, but have each student's questions appear in a different order. If there are 128 students in the grade 12 class, what is the least number of questions the quiz must contain so everyone gets a test with the questions in a different order.

ANS:
Assume there are two questions on the test. They could appear in 2! possible ways. This can be carried on for additional questions.
For 3 questions, there are 3! = 6 possible orders.
For 4 questions, there are 4! = 24 possible orders.
For 5 questions, there are 5! = 120 possible orders.
For 6 questions, there are 6! = 720 possible orders.

There must be 6 questions for everyone to get a test with the questions in a different order.

DIF: Average OBJ: Section 11.1 NAT: PC1 TOP: Permutations
KEY: fundamental counting principle

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10. Solve for n: $n({}_{10}P_2) = {}_{10}C_8$.

ANS:

$$n\,{}_{10}P_2 = {}_{10}C_8$$

$$n\left(\frac{10!}{8!}\right) = \frac{10!}{2!8!}$$

$$n = \frac{10!8!}{10!2!8!}$$

$$n = \frac{1}{2!}$$

$$n = \frac{1}{2}$$

DIF: Average OBJ: Section 11.1 | Section 11.2 NAT: PC1 | PC2 | PC3
TOP: Permutations | Combinations KEY: factorial | permutations | combinations

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PROBLEM

1. A bingo card has 5 columns. The tops of the columns are labelled with the letters B, I, N, G, and O. Each column contains exactly five numbers, except for the N column, which has a free space in the centre. No numbers repeat on the card. The B column contains numbers from 1 to 15, the I column from 16 to 30, the N column from 31 to 45, the G column from 46 to 60, and the O column from 61 to 75.

a) How many unique bingo cards can be printed?

b) If every Canadian were able to count one card per second, how many years would it take to count all of the cards? Assume the Canadian population is about 34 000 000.

c) Suppose the cards are printed on card stock 0.3 mm thick. How many kilometres high would a stack of all of the possible cards be?

ANS:

a) For columns B, I, G, and O, there are 15(14)(13)(12)(11) or 360 360 possible choices. For the N column, there are 15(14)(13)(12) or 32 760 choices.

The total number of choices is $(360\ 360)^4(32\ 760)$ or about $5.52 \infty 10^{26}$.

b) $\dfrac{5.52 \times 10^{26}\ \text{cards}}{34\ 000\ 000\ \text{cards/s}} \approx 1.62 \times 10^{19}\ \text{s}$

$$= (1.62 \times 10^{19}\ \text{s})\left(\frac{1\ \text{h}}{3600\ \text{s}}\right)\left(\frac{1\ \text{day}}{24\ \text{h}}\right)\left(\frac{1\ \text{year}}{365\ \text{days}}\right)$$

$$\approx 5.14 \times 10^{11}\ \text{years}$$

It would take about $5.14 \infty 10^{11}$ years to count all of the cards.

c) $(5.52 \times 10^{26}\ \text{cards})(0.3\ \text{mm/card}) \approx 1.66 \times 10^{26}\ \text{mm}$

$$= 1.66 \times 10^{26}\ \text{mm}\left(\frac{1\ \text{km}}{1 \times 10^{6}\ \text{mm}}\right)$$

$$= 1.66 \times 10^{20}\ \text{km}$$

The stack would be $1.66 \infty 10^{20}$ km high.

DIF: Average OBJ: Section 11.1 NAT: PC1 TOP: Permutations
KEY: fundamental counting principle

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2. A standard deck of playing cards contains 4 suits (spades, clubs, diamonds, and hearts), each with 13 cards.
a) How many different 5-card hands are possible?
b) How many different 5-card hands with only black cards are possible?
c) How many different 5-card hands are possible containing at least 3 black cards?

ANS:
a) ${}_{52}C_5 = 2\,598\,960$
b) There are 26 black cards (spades and clubs), so there are ${}_{26}C_5$ or 65 780 possible hands with only black cards.
c) There are 3 possible situations:
Case 1: all 5 cards are black:

$${}_{26}C_5 = 65\,780$$

Case 2: 4 black cards, 1 red card

$$P(\text{black}) \times P(\text{red}) = ({}_{26}C_4)({}_{26}C_1)$$
$$= 388\,700$$

Case 3: 3 black cards, 2 red card

$$P(\text{black}) \times P(\text{red}) = ({}_{26}C_3)({}_{26}C_2)$$
$$= 845\,000$$

The total number of different 5-card hands containing at least 3 black cards is
65 780 + 388 700 + 845 000 or 1 299 480.

DIF: Difficult OBJ: Section 11.2 NAT: PC3 TOP: Combinations
KEY: combinations

3. Expand and simplify: $(x+y)^6 - (x-y)^6$.

ANS:
$(x+y)^6 - (x-y)^6$
$= x^6 + 6x^5y^1 + 15x^4y^2 + 20x^3y^3 + 15x^2y^4 + 6x^1y^5 + y^6 - (x^6 - 6x^5y^1 + 15x^4y^2 - 20x^3y^3 + 15x^2y^4 - 6x^1y^5 + y^6)$
$= 12x^5y + 40x^3y^3 + 12xy^5$
or
$(x+y)^6 - (x-y)^6$
$= [(x+y)^3 + (x-y)^3][(x+y)^3 - (x-y)^3]$
$= (x^3 + 3x^2y + 3xy^2 + y^3 + x^3 - 3x^2y + 3xy^2 - y^3)[x^3 + 3x^2y + 3xy^2 + y^3 - (x^3 - 3x^2y + 3xy^2 - y^3)]$
$= (2x^3 + 6xy^2)(6x^2y + 2y^3)$
$= 12x^5y + 4x^3y^3 + 36x^3y^3 + 12xy^5$
$= 12x^5y + 40x^3y^3 + 12xy^5$

DIF: Average OBJ: Section 11.3 NAT: PC4 TOP: The Binomial Theorem
KEY: binomial expansion | binomial theorem

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4. The number of different permutations using all of the 1-digit numbers of a set is given by $\frac{7!}{3!2!}$.

a) What is the smallest number that can be created that meets these conditions? Explain your reasoning.
b) What is the difference between the largest number and the smallest number? Explain your reasoning.

ANS:
a) The single digits range from 0 to 9. One digit is used 3 times, one digit is used twice, and the rest are used only once. Since there are seven digits in total, the smallest digits make the smallest number. Therefore, the number is 1 000 123.
b) The largest number can be found in a similar way using the largest possible digits. The largest number is 9 998 876. The difference between the numbers is 9 998 876 – 1 000 123 or 8 998 753.

DIF: Difficult OBJ: Section 11.1 NAT: PC2 TOP: Permutations
KEY: permutations

5. Arrange the following provincial capitals from the least to the greatest number of arrangements of letters: Fredericton, Charlottetown, Edmonton, Victoria, and Winnipeg.

ANS:
Winnipeg and Edmonton have the same number of combinations:
Winnipeg: $\frac{8!}{2!2!} = 10\,080$
Edmonton: $\frac{8!}{2!2!} = 10\,080$
Victoria: $\frac{8!}{2!} = 20\,160$
Fredericton: $\frac{11!}{2!2!} = 9\,979\,200$
Charlottetown: $\frac{13!}{3!2!} = 518\,918\,400$

DIF: Average OBJ: Section 11.1 NAT: PC1 TOP: Permutations
KEY: factorial | permutations

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6. Matthew buys a combination lock that uses letters instead of numbers. There are five tumblers on the lock, with the letters from A to T on each tumbler.
a) What is the probability that Matthew chooses WATER as the combination?
b) How many possible combinations are there if the letters can be repeated?
c) If it takes 12 s to test one combination, how long will it take to check all of the combinations in part b) if Matthew forgets his combination? Express your answer in years and months.

ANS:
a) Since W is not one of the letters on the tumblers, the probability is 0.
b) $20^5 = 3\ 200\ 000$
c) It will take (3 200 000)(12) or 38 400 000 s to check all of the combinations. This is about 1 year and 2.6 months.

DIF: Difficult OBJ: Section 11.1 NAT: PC1 TOP: Permutations
KEY: permutations

7. The chorus of a play has 17 females and 13 males. The director wishes to meet with 6 of them to discuss the upcoming production.
a) How many selections are possible?
b) How many selections are possible if the group consists of three females and three males?
c) One of the male students is named Ajay. How many six-member selections consisting of Ajay, two other males, and three females are possible?

ANS:
a) $${}_{30}C_6 = \frac{30!}{(30-6)!6!}$$

$$= \frac{30!}{24!6!}$$

$$= 593\ 775$$

There are 593 775 possible ways of selecting 6 members of the chorus.
b) $$({}_{17}C_3)({}_{13}C_3) = \left(\frac{17!}{(17-3)!3!}\right)\left(\frac{13!}{(13-3)!3!}\right)$$

$$= 680(286)$$

$$= 194\ 480$$

There are 194 480 possible ways of selecting three females and three males.
c) There is only one way of selecting Ajay. This leaves 12 males left to choose the other two males.
$(1)({}_{12}C_2)({}_{17}C_3) = 66(680)$

$$= 44\ 880$$

There are 44 880 possible ways to select the group that includes Ajay.

DIF: Difficult OBJ: Section 11.2 NAT: PC3 TOP: Combinations
KEY: combinations

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8. A group of police officers are planning a bike ride in Alberta to raise money for cancer research. They are planning to go to three areas of the province. They want to minimize their travel between areas with possible visits to each of the following communities:

Edmonton Area	**Calgary Area**	**Lethbridge Area**
Edmonton	Calgary	Lethbridge
St. Albert	Cochrane	Coalhurst
Spruce Grove	Cranston	Coaldale
Stony Plain	Chestermere	
Sherwood Park		

a) If the group goes to 7 communities in total, how many different schedules can be created if there are no restrictions on which communities they visit in each area?
b) If the group must visit all of the communities in the Edmonton Area, 3 in the Calgary area, and 2 in the Lethbridge area, how many different schedules can be created?
c) If the group decides that Sherwood Park and Coaldale should not be on the schedule due to distance, how many different schedules can be created if the group visits all of the communities in the Edmonton area, 2 in the Calgary area, and 2 in the Lethbridge area?

ANS:
a) Order will be important in the schedule due to the distances involved. There are 12 possible places to visit:

$$_{12}P_7 = \frac{12!}{(12-7)!}$$

$$= 3\,991\,680$$

There are 3 991 680 possible ways to organize the schedule.
b) The communities can be organized in the following ways:
Edmonton area: $_5P_5 = 120$ ways
Calgary area: $_4P_3 = 24$ ways
Lethbridge: $_3P_2 = 6$ ways
The total number of schedules is $120 \infty 24 \infty 6$ or 17 280.
c) Removing Sherwood Park and Coaldale from the schedule changes the number of possible schedules:

$$(_4P_4)(_4P_2)(_2P_2) = (24)(12)(2)$$

$$= 576$$

There are 576 possible schedules.

DIF: Average OBJ: Section 11.2 NAT: PC3 TOP: Permutations
KEY: permutations

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9. To win the grand prize in lottery A, a player must select all six of the winning numbers drawn from the numbers 1 to 49. To win in lottery B, a player must select all seven of the winning numbers drawn from 1 to 49. Bernadette argues that the chances of randomly selecting the winning number for lottery A are seven times as good as winning for lottery B. Create an argument to agree or disagree with this statement.

ANS:
The total number of possible tickets for lottery A is ${}_{49}C_6$ or 13 983 816.
Therefore, the probability of randomly selecting the winning numbers is

$$P(\text{A}) = \frac{1}{13\,983\,816}$$

The total number of possible tickets for lottery B is ${}_{49}C_7$ or 85 900 584.
Therefore, the probability of randomly selecting the winning numbers is

$$P(\text{B}) = \frac{1}{85\,900\,584}$$

Compare the probabilities:

$$\frac{P(\text{A})}{P(\text{B})} = \frac{\frac{1}{13\,983\,816}}{\frac{1}{85\,900\,584}}$$

$$= \frac{85\,900\,584}{13\,983\,816}$$

$$\approx 6.14$$

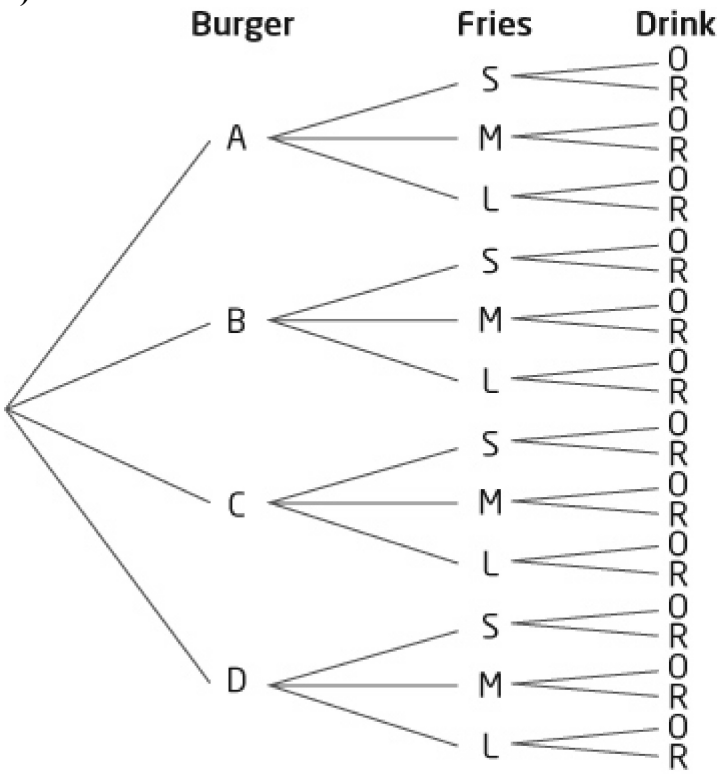
Bernadette is not correct. The probability of winning lottery A is only 6.14 times as great.

DIF: Difficult OBJ: Section 11.2 NAT: PC3 TOP: Combinations
KEY: combinations

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10. Tanya goes to a fast food stand at the beach. There are 4 types of burgers, 3 sizes of French fries, and either orange pop or root beer to drink.
a) Create a tree diagram to show the possible choices of lunch if one of each item can be selected.
b) In how many ways can Tanya buy 2 burgers, 2 fries, and 2 drinks for her and her friend?
c) If Tanya does not like root beer and her friend does not like orange pop, how many possible choices are there?

ANS:
a)

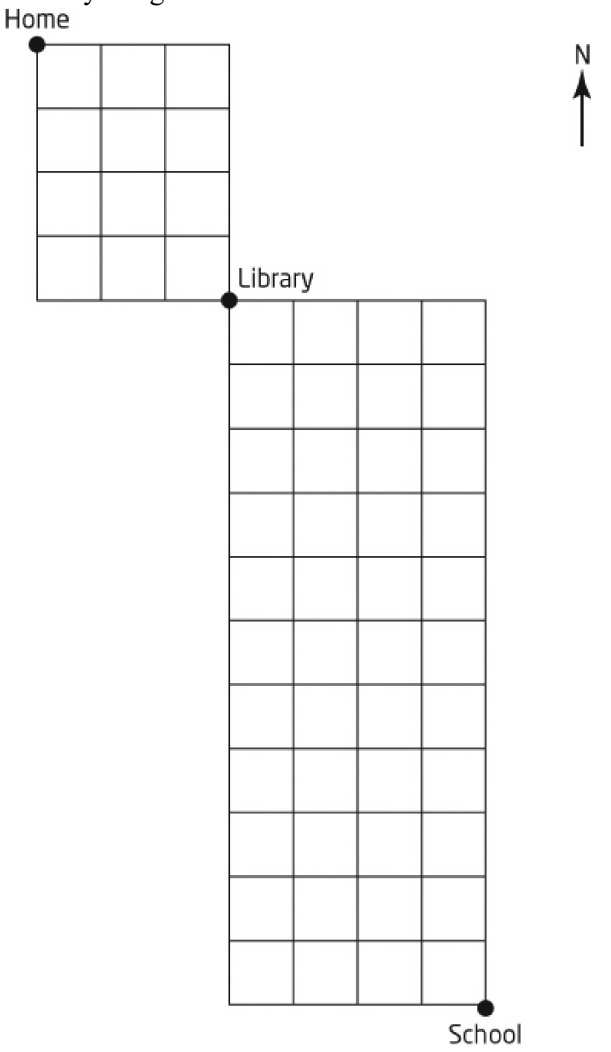


There are 24 possible choices for lunch.
b) The total numbers of ways is (24)(24) = 576.
c) Tanya and her friend will both have half of their choices eliminated. The total number of choices is (12)(12) or 144.

DIF: Easy OBJ: Section 11.1 NAT: PC1 TOP: Permutations
KEY: tree diagram

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11. On a Saturday, Charlie has to go to the library to study for a few hours, and then to the school to play a volleyball game.



a) How many routes are there for Charlie to go from home to the library if she only moves south and east?
b) How many routes are there for her to go from home to school moving only south and east?
c) Assuming Charlie moves south and east going from home to school and north and west going from school to home, how many routes are there for her to complete the round trip?
d) If Charlie could walk each route from home to school in 40 min, how long would it take her and her 22 classmates to walk all of the routes? Consider only the route from home to school, not the round trip.

ANS:

a) $\frac{7!}{3!4!} = \frac{5040}{144}$

$= 35$

There are 35 possible routes from home to the library.

b) $\left(\frac{7!}{3!4!}\right)\left(\frac{15!}{4!11!}\right) = (35)(1365)$

$= 47\,775$

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There are 47 775 possible routes from home to school.
c) There are 47 775 possible routes in each direction, so the total number of routes from home to school and back is $47\,775^2$ or 2 282 450 625.
d) The total time to walk all of the routes for 1 person is (40)(47 775) or 1 911 000 min.
Charlie and her classmates (23 people in all) can walk the routes in 1 911 000 ÷ 23 or about 83 087.0 min.
Convert to days:

$$\frac{83\,087.0}{(60)(24)} \approx 57.7$$

It would take about 58 days for the 23 students to walk all of the routes from home to school.

DIF: Difficult OBJ: Section 11.1 NAT: PC1 | PC2 TOP: Permutations
KEY: permutations | fundamental counting principle

12. Ms. Jackson's class of 28 students all submitted projects for a recent math fair. The principal asked Ms. Jackson to prepare a display for the upcoming parent–teacher evening.
a) In how many ways can 20 of the projects be chosen?
b) The principal asks to set the 4 regional winners out of the 20 selected near the entrance of the room. In how many ways can the projects be displayed? Leave your answer in factorial form.

ANS:
a) $${}_{28}C_{20} = \frac{28!}{20!8!}$$

$$= 3\,108\,105$$

There are 3 108 105 possible ways to choose the 20 projects.
b) There are 4! ways to arrange the 4 regional winners, and 16! ways to arrange the other 16. Thus, there are (4!)(16!) ways to display the projects.

DIF: Average OBJ: Section 11.1 NAT: PC2 TOP: Permutations
KEY: permutations

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13. Mr. Williams gives his class a test as part of their final mark for a course. Mr. Williams believes in choice and offers some options to students. How many different options are possible if
a) students must complete three of the six questions in the first part and two of the three questions in the second part?
b) students must answer three of the questions in the first part and at least two of the three questions in the second part?

ANS:
a) The number of ways of selecting the questions in the first part is ${}_6C_3$ or 20.
The number of ways of selecting the questions in the second part is ${}_3C_2$ or 3.
The number of possible question selections is (20)(3) or 60.
b) Since the students must answer at least 2 of the questions in the second part, they can answer two or three questions.
Case 1: Answer two questions in the second part
$({}_6C_3)({}_3C_2) = 20(3)$

$= 60$

Case 2: Answer three questions in the second part
$({}_6C_3)({}_3C_3) = 20(1)$

$= 20$

The number of ways of choosing three of the questions in the first part and at least two questions in the second part is 60 + 20 or 80.

DIF: Difficult OBJ: Section 11.2 NAT: PC3 TOP: Combinations
KEY: combinations | cases

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14. Seven students (Catherine, Jim, Jerome, Lucia, Lisa, Melinda, and Zyoji) are entered in a debate contest.

a) In how many ways can they give their arguments if the following conditions must be met?

i) Lucia must go second.

ii) Lisa must go first and Zyoji must go second to last.

iii) All of the boys go first followed by the girls.

iv) All of the girls go first followed by the boys.

v) Females and males alternate with a female going first.

b) Are there any generalizations you can make from the solutions to part a)i) to v)?

ANS:

a) i) Assume the order is ____, Lucia, ____, ____, ____, ____, ____, where the only spot taken is the second position by Lucia. The number of choices is determined by the number of students remaining for each position:

6, Lucia (1), 5, 4, 3, 2, 1

Thus, there are 6(5)(4)(3)(2)(1) or 720 ways to organize the students.

ii) Assume the order is Lisa, ____, ____, ____, ____, Zyoji, ____, where the only spots taken are the first position by Lisa and the second-last position by Zyoji. The number of choices is determined by the number of students remaining for each position:

Lisa (1), 5, 4, 3, 2, Zyoji (1), 1

Thus, there are 120 ways to organize the students.

iii) There are (3)(2)(1)(4)(3)(2)(1) or 144 different ways if all the boys go first.

iv) There are (4)(3)(2)(1)(3)(2)(1) or 144 different ways if all the girls go first

v) There are (4)(3)(3)(2)(2)(1)(1) or 144 different ways if boys and girls alternate.

b) When determining order, the fundamental counting principal can be used. Any item that must be in a specific position has only one option. All other position have as many options as there are choices that meet the criteria (e.g., boys, girls).

DIF: Average OBJ: Section 11.1 NAT: PC1 TOP: Permutations
KEY: fundamental counting principle | conditions

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15. Simplify.

a) $\frac{{}_{500}C_{400}}{{}_{500}C_{100}}$

b) $\left(\frac{{}_{500}P_{400}}{{}_{500}P_{100}}\right)\left(\frac{1}{{}_{400}P_{300}}\right)$

ANS:

a) $\frac{{}_{500}C_{400}}{{}_{500}C_{100}} = \frac{\frac{500!}{(500-400)!400!}}{\frac{500!}{(500-100)!100!}}$

$= \left(\frac{500!}{(500-400)!400!}\right)\left(\frac{(500-100)!100!}{500!}\right)$

$= \left(\frac{500!}{100!400!}\right)\left(\frac{400!100!}{500!}\right)$

$= 1$

b) $\left(\frac{{}_{500}P_{400}}{{}_{500}P_{100}}\right)\left(\frac{1}{{}_{400}P_{300}}\right) = \left(\frac{\frac{500!}{(500-400)!}}{\frac{500!}{(500-100)!}}\right)\left(\frac{1}{\frac{400!}{(400-300)!}}\right)$

$= \left(\frac{500!}{100!}\right)\left(\frac{400!}{500!}\right)\left(\frac{100!}{400!}\right)$

$= 1$

DIF: Difficult OBJ: Section 11.1 | Section 11.2 NAT: PC2 | PC3
TOP: Permutations | Combinations KEY: factorial | permutations | combinations

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16. You have been asked to determine the probability of prizes for a new lottery. The lottery will be called Lotto 5-45.To win a prize, a player must pick numbers from 1 to 45. The number of correct numbers chosen determines the amount won.

Correct Numbers	Prize	Probability of Winning
5	Jackpot	
4	$1000	
3	$10	
2	Free Ticket	

a) Complete the table.

b) To win a lottery jackpot in Lotto Fab, a player must chose six numbers from 1 to 49 without repetition. Are the chances of winning the jackpot in Lotto 5-45 better than in Lotto Fab?

c) In Lotto Fab, a player also has to choose two correct numbers to win a free ticket. Which lottery has the better chance of winning a free ticket?

ANS:

a) For each of the prizes the probabilities is calculated using

$$\frac{\text{favourable cases}}{\text{total cases}} = \frac{({}_{\text{Winning numbers}}C_{\text{number picked}})({}_{\text{Non-winning numbers}}C_{\text{number picked}})}{{}_{45}C_5}$$

$$= \frac{({}_{\text{Winning numbers}}C_{\text{number picked}})({}_{\text{Non-winning numbers}}C_{\text{number picked}})}{1\,221\,759}$$

Correct Numbers	Prize	Probability of Winning
6	Jackpot	$\frac{({}_{\text{Winning numbers}}C_{\text{number picked}})({}_{\text{Non-winning numbers}}C_{\text{number picked}})}{1\,221\,759} = \frac{({}_5C_5)({}_{40}C_0)}{1\,221\,759} = \frac{1}{1\,221\,759}$
4	$1,000	$\frac{({}_{\text{Winning numbers}}C_{\text{number picked}})({}_{\text{Non-winning numbers}}C_{\text{number picked}})}{1\,221\,759} = \frac{({}_5C_4)({}_{40}C_1)}{1\,221\,759} = \frac{200}{1\,221\,759}$
3	$10	$\frac{({}_{\text{Winning numbers}}C_{\text{number picked}})({}_{\text{Non-winning numbers}}C_{\text{number picked}})}{1\,221\,759} = \frac{({}_5C_3)({}_{40}C_2)}{1\,221\,759} = \frac{7800}{1\,221\,759}$
2	Free Ticket	$\frac{({}_{\text{Winning numbers}}C_{\text{number picked}})({}_{\text{Non-winning numbers}}C_{\text{number picked}})}{1\,221\,759} = \frac{({}_5C_2)({}_{40}C_3)}{1\,221\,759} = \frac{98\,800}{1\,221\,759}$

b) The probability of winning the jackpot for Lotto Fab is $\frac{1}{{}_{49}C_6}$ or $\frac{1}{13\,983\,816}$.

Since this is less than $\frac{1}{1\,221\,759}$, the probability of winning Lotto 5-45 is greater.

c) The probability of winning a free ticket with Lotto Fab is

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$$\frac{({}_{\text{Winning numbers}}C_{\text{number picked}})({}_{\text{Non-winning numbers}}C_{\text{number picked}})}{13\,983\,816} = \frac{({}_6C_2)({}_{43}C_4)}{13\,983\,816}$$

$$= \frac{1\,851\,150}{13\,983\,816}$$

$$\approx 0.13$$

The probability of winning a free ticket with Lotto 5/45 is

$$\frac{({}_{\text{Winning numbers}}C_{\text{number picked}})({}_{\text{Non-winning numbers}}C_{\text{number picked}})}{1\,221\,759} = \frac{({}_5C_2)({}_{40}C_3)}{1\,221\,759}$$

$$= \frac{98\,800}{1\,221\,759}$$

$$\approx 0.08$$

The probability of winning a free ticket is better with Lotto Fab.

DIF: Difficult + OBJ: Section 11.2 NAT: PC3 TOP: Combinations
KEY: combinations

17. Expand using the binomial theorem: $\left(x^2 - \frac{x}{2}\right)^5$.

ANS:

$$\left(x^2 - \frac{x}{2}\right)^5$$

$$= {}_5C_0(x^2)^5\left(-\frac{x}{2}\right)^0 + {}_5C_1(x^2)^4\left(-\frac{x}{2}\right)^1 + {}_5C_2(x^2)^3\left(-\frac{x}{2}\right)^2 + {}_5C_3(x^2)^2\left(-\frac{x}{2}\right)^3 + {}_5C_4(x^2)^1\left(-\frac{x}{2}\right)^4 + {}_5C_5(x^2)^0\left(-\frac{x}{2}\right)^5$$

$$= x^{10} - \frac{5}{2}x^9 + \frac{5}{2}x^8 - \frac{5}{4}x^7 + \frac{5}{16}x^6 - \frac{x^5}{32}$$

DIF: Average OBJ: Section 11.3 NAT: PC4 TOP: The Binomial Theorem
KEY: binomial expansion | binomial theorem

18. Express $16x^4 - 96x^3y + 216x^2y^2 - 216xy^3 + 81y^4$ in the form $(ax + by)^n$.

ANS:

The first term is of the form ${}_nC_0(ax)^n$, or a^nx^n. Since the first term of the expanded binomial is $16x^4$, $a = 2$ and $n = 4$. The last term is in the form ${}_nC_n(by)^n$, or b^ny^n. Since the last term of the expanded binomial is $81y^4$, and the signs alternate between + and –, $b = -3$.

Thus, $16x^4 - 96x^3y + 216x^2y^2 - 216xy^3 + 81y^4 = (2x - 3y)^4$.

DIF: Difficult + OBJ: Section 11.3 NAT: PC4 TOP: The Binomial Theorem
KEY: binomial expansion | binomial theorem

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19. Which of the following binomial expansions has the greater coefficient: the 12th term in the expansion of $(2x-2)^{13}$ or the 8th term in the expansion of $(3x-2)^{10}$?

ANS:

12th term in expansion of $(2x-2)^{13}$:

$${}_{13}C_{11}(2x)^{13-11}(-2)^{11} = 78(2x)^2(-2048)$$

$$= -638\,976x^2$$

8th term in expansion of $(3x-2)^{10}$:

$${}_{10}C_7(3x)^{10-7}(-2)^7 = 120(3x)^3(-128)$$

$$= -414\,720x^3$$

Since $-414\,720 > -638\,976$, the 8th term in the expansion of $(3x-2)^{10}$ has the greater coefficient.

DIF: Difficult OBJ: Section 11.3 NAT: PC4 TOP: The Binomial Theorem
KEY: binomial expansion | binomial theorem

20. **a)** If all of the letters in the word ROCKING are used, how many arrangements can be made with all of the vowels together?
b) Which word has the greater number of different arrangements: CATCHPHRASE or MISSISSIPPI?
c) Is it possible to create a word whose letters can be arranged with exactly 20 possible different arrangements? If so, explain how.

ANS:

a) There are 2 vowels, so they can be considered a single unit since they must be together. These can be arranged in 2! = 2 distinct ways. Therefore, the total number of arrangements is (6!)(2!) or 1440.

b) CATCHPHRASE: $\frac{11!}{2!2!2!} = 4\,989\,600$ MISSISSIPPI: $\frac{11!}{4!4!2!} = 34\,650$

c) Yes. Examples may vary. Sample answer:
A 5-letter word with 5 unique letters will result in 5! = 120 possible combinations.
If 1 letter repeats 3 times, there are $\frac{5!}{3!}$ or 20 possible arrangements
A 6-letter word with 2 letters repeating 3 times has $\frac{6!}{3!3!}$ or 20 possible arrangements.

DIF: Average OBJ: Section 11.1 NAT: PC1 TOP: Permutations
KEY: permutations

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